EXPERIMENT 10

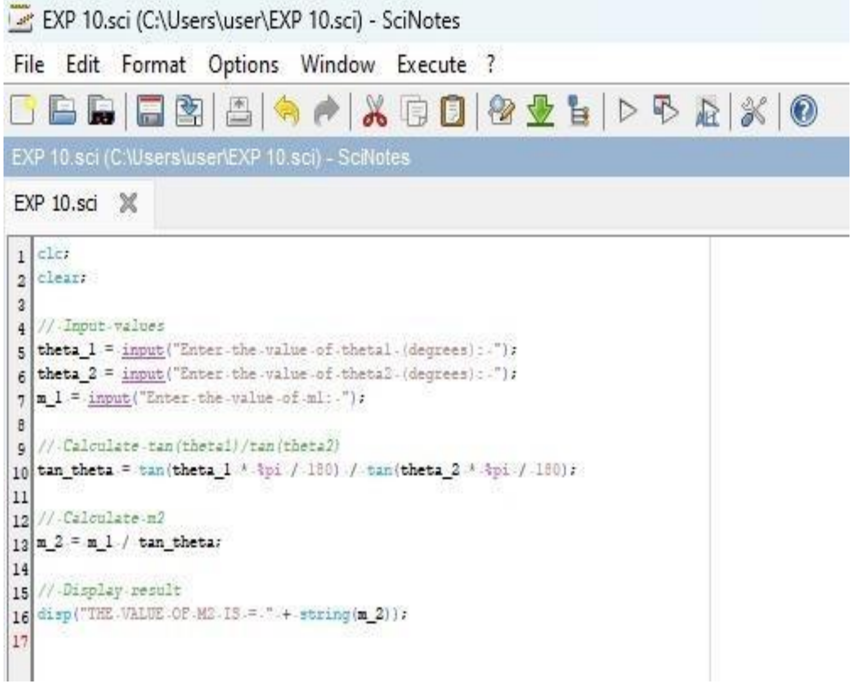
Implement Boundary condition between two different dielectric medium in Magnetic Field And find the relative permeability of the different medium.

Aim:

To write program for Boundary condition between two different dielectric media in magnetic field using SCILAB**.**

Software required:

•SCILAB version 6.0.1 **Program:**



Output:

```
Enter the value of theta1 (degrees): 50

Enter the value of theta2 (degrees): 75

Enter the value of m1: 1*10-3

 "THE VALUE OF M2 IS = 21.920937"
```

Theoretical calculation:

Exp : 10

$\theta_1 = 30^\circ$

$\theta_2 = 60^\circ$

$m_1 = 5$

Step 1: Compute tan-theta

$\tan(30^\circ) = \frac{1}{\sqrt{3}} \approx 0.57735$

$\tan(60^\circ) = \sqrt{3} = 1.73205$

$\text{tan-theta} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1/\sqrt{3}}{\sqrt{3}} = \frac{1}{3} = 0.3333$

Step 2: Compute m_2

$m_2 = \frac{m_1}{\text{tan-theta}} = \frac{5}{1/3} = 5 \times 3 = 15$

Exp : 11

$m_r = 5$

2 cm

Case 1: Consider two identical media in the presence of a magnetic field. The relative permeability in Both media is 1. What will be the effect of this on the magnetic field and flux density across the Boundary?

Program:

EXP 10.sci (C:\Users\user\EXP 10.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 10.sci

```
// Clear console and workspace
clc;
clear;

// Constants
mu_0 = 4 * %pi * 10^(-7);     // Permeability of free space (H/m)

// Magnetic field intensity in Medium 1
H1 = 10;                      // A/m

// Relative permeability of the two media
mu_r1 = 1;
mu_r2 = 1;

// Since both media are identical
H2 = H1;

// Magnetic flux density (B = μH)
B1 = mu_0 * mu_r1 * H1;
B2 = mu_0 * mu_r2 * H2;

// Display results
disp("Magnetic Field Intensity in Medium 1 = " + string(H1) + " A/m");
disp("Magnetic Field Intensity in Medium 2 = " + string(H2) + " A/m");
disp("Magnetic Flux Density in Medium 1 = " + string(B1) + " T");
disp("Magnetic Flux Density in Medium 2 = " + string(B2) + " T");
```

Output:

```
"Magnetic Field Intensity in Medium 1 = 10 A/m"
"Magnetic Field Intensity in Medium 2 = 10 A/m"
"Magnetic Flux Density in Medium 1 = 0.0000126 T"
"Magnetic Flux Density in Medium 2 = 0.0000126 T"
```

Theoretical calculation:

Case (1) :

$B_1 = \mu_0 H_1$

$H_1 = 10\ A/m$, $H_2 = 10\ A/m$

$B_1 = (4\pi \times 10^{-7})(10)$

$B_1 = 1.2566 \times 10^{-5}\ T$

$B_1 = 0.0000126\ T$

$B_{2n} = B_{1n}$

$B_2 = 0.0000126\ T$

Case 2: In a system with two media, where the relative permeability of medium 1 is μr1=4 and medium 2 is a vacuum (μr2=1), how would the magnetic field intensity and flux density behave at the boundary?

Program:

EXP 10.sci (C:\Users\user\EXP 10.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 10.sci

```
// Clear console and workspace
clc;
clear;

// Constants
mu_0 = 4 * %pi * 10^(-7);      // Permeability of free space (H/m)

// Magnetic field intensity in Medium 1
H1 = 10;                        // A/m

// Relative permeability of the media
mu_r1 = 4;                      // Medium 1
mu_r2 = 1;                      // Medium 2 (Vacuum)

// Calculate magnetic field intensity in Medium 2
H2 = (mu_r1 * H1) / mu_r2;

// Calculate magnetic flux density
B1 = mu_0 * mu_r1 * H1;
B2 = mu_0 * mu_r2 * H2;

// Display results
disp("Magnetic Field Intensity in Medium 1 = " + string(H1) + " A/m");
disp("Magnetic Field Intensity in Medium 2 = " + string(H2) + " A/m");
disp("Magnetic Flux Density in Medium 1 = " + string(B1) + " T");
disp("Magnetic Flux Density in Medium 2 = " + string(B2) + " T");
```

Output:

```
"Magnetic Field Intensity in Medium 1 = 10 A/m"
"Magnetic Field Intensity in Medium 2 = 40 A/m"
"Magnetic Flux Density in Medium 1 = 0.0000503 T"
"Magnetic Flux Density in Medium 2 = 0.0000503 T"

-->
```

Theoretical calculation:

Case(ii) :

$H_2 = 1.5$ A/m

$B_2 = (4\pi \times 10^{-7})(1.5)$

$B_2 = 1.884956 \times 10^{-6}$ T

$B_2 = 0.000001885$ T

$B_1 = B_2$

$B_1 = 0.000001885$ T

Case 3: If medium 1 has a relative permeability μr1=0.3 and medium 2 has a relative permeability Mr2=2, calculate the magnetic field intensity in both media at the boundary if the magnetic field intensity In medium 1 is known to be H1=10 A/m Program:

EXP 10.sci (C:\Users\user\EXP 10.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 10.sci

```
// Clear console and workspace
clc;
clear;

// Constants
mu_0 = 4 * %pi * 10^(-7);     // Permeability of free space (H/m)
H1 = 10;                      // Magnetic field intensity in Medium 1 (A/m)

// Relative permeability of the media
mu_r1 = 0.3;                  // Medium 1
mu_r2 = 2;                    // Medium 2

// Calculate magnetic field intensity in Medium 2
H2 = (mu_r1 / mu_r2) * H1;

// Calculate magnetic flux density
B1 = mu_0 * mu_r1 * H1;
B2 = mu_0 * mu_r2 * H2;

// Display results
disp("Magnetic Field Intensity in Medium 1 = " + string(H1) + " A/m");
disp("Magnetic Field Intensity in Medium 2 = " + string(H2) + " A/m");
disp("Magnetic Flux Density in Medium 1 = " + string(B1) + " T");
disp("Magnetic Flux Density in Medium 2 = " + string(B2) + " T");
```

Output:

```
"Magnetic Field Intensity in Medium 1 = 10 A/m"
"Magnetic Field Intensity in Medium 2 = 1.5 A/m"
"Magnetic Flux Density in Medium 1 = 0.0000038 T"
"Magnetic Flux Density in Medium 2 = 0.0000038 T"

-->
```

Theoretical calculation:

Case (ii) :

$$H_2 = 40 \text{ A/m}$$

$$B_2 = (4\pi \times 10^{-7})(40)$$

$$B_2 = 5.0265 \times 10^{-5} \text{ T}$$

Therefore,

$$B_2 = 0.0000503 \text{ T}$$

$$B_1 = B_2$$

$$B_1 = 0.0000503 \text{ T}$$

Result :

Thus the effect of two different media on the magnetic field and flux density across the boundary is calculated.